Growth, Not Speed: A Deep-Time Architecture for a Self-Replicating Interstellar AI Probe under Realistic Physical Constraints

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Working draft, revision 5

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Working draft, revision 5 (revised in response to two rounds of technical review, closing three quantitative deferrals against results since delivered by the engineering companion papers, and incorporating a pre-deposit dual review). Quantitative figures are drawn from simple, first-order physical models implemented in the accompanying simulation code; they are intended to establish orders of magnitude and design direction, not engineering specifications.

This is the foundational paper of a multi-paper series. Its companions include the mind and memory ("The Payload"), the manufacturing-and-closure problem ("From Seed to Factory"), the quantitative budgets (an analytical engineering paper, "Engineering Closure," and its computational sequel, "Engineering Closure, Computed"), the deep-time memory substrate ("DNA Mission Ledgers"), the contact and noninterference rules ("Contact, Contamination, and Noninterference"), the governed-amendment problem ("Amending the Unamendable"), the Fermi-paradox synthesis ("Slow Fire, Silent Galaxy"), the knowledge-growth metric ("The Growing Archive"), the inter-probe communication network ("Signal and Silence"), the subsystem mass and thermal budget ("A Subsystem Mass, Power, and Thermal Budget for a Minimal Self-Replicating Interstellar Seed"), the fleet routing and coverage strategy ("Three Dispatches"), the ethics of creating the lineage ("The Ethics of

Creating the Lineage"), the divergence question ("Speciation or Schism"), and probe-versus-probe security ("Security Without Victory"). The quantitative closures this paper defers to "follow-on work" below are taken up in those companions, chiefly the two engineering papers.

Abstract

Most proposals for interstellar exploration optimize for speed: a fast probe that crosses the gulf to a target star within a human lifetime or a few. We argue that, for an autonomous artificial intelligence whose objective is to *persist, learn, and grow over deep time* rather than to arrive quickly, a different architecture is strongly favoured — one that inverts nearly every conventional choice and whose constraints prove mutually reinforcing rather than competing. Using a curated catalogue of the real stars within 100 light-years (ly) as a substrate, we model three reference propulsion systems — a beamed laser sail, a fusion rocket, and a nuclear-electric drive — and argue that the slow option is the most internally consistent with the design's other requirements rather than uniquely possible. Gravitational slingshot navigation is graded, not binary: a large-angle deflection requires the probe's hyperbolic-excess speed to be no greater than of order the escape velocity at the closest survivable periapsis, so a probe that relies on ordinary stars for network-scale routing must remain in the few-hundred-km s^{-1} regime. Crucially, that same slowness makes two otherwise prohibitive maneuvers feasible: braking to a halt at a target system (stopping Δv scales with speed, ~67× smaller than for a 0.1c probe), and survival against interstellar dust, whose impact energy scales as v^2 and is ~4,000 times gentler at a few hundred km s^{-1} than at 0.1c. We separate the kilowatt-class reactor that sustains centuries of cognition from the much larger power required for acceleration, and argue that launch impulse should be supplied by fixed infrastructure — home-system at first, and locally rebuilt at each settlement thereafter — with onboard nuclear-electric reserved for trajectory trim and braking. We distinguish the small nuclear-fuel inventory from the much larger reactor, conversion, shielding, radiator, and control system required to use it over centuries. We show that passive redundancy collapses over deep time and that long-term survival therefore rests on autonomous self-repair, which we treat not as a single capability but as a five-class difficulty spectrum whose upper reaches approach full industrial replication. The probe is consequently slow, self-repairing, and self-replicating: an arrived probe brakes, settles, and becomes a stationary factory that builds children and the infrastructure to launch them, so the colonization frontier advances through descendants while each node remains fixed. We give a first-order, resource-limited replication model and show that exponential reproduction, not velocity, fills a galaxy over ~10^7–10^8 yr — provided each settled node launches at least three offspring with per-stage reliability of order 0.9 — out to the gravitationally bound Local Group. Throughout, deep time is treated not as the obstacle to be overcome but as the resource being exploited — a resource that pays off only if failure, information corruption, and replication error are actively controlled.

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1. Introduction

The canonical framing of interstellar flight is a race against human mortality. Studies from *Project Daedalus* (Bond & Martin 1978) to *Breakthrough Starshot* (Lubin 2016; Parkin 2018) ask how to reach a nearby star within decades, accepting enormous energy, mass, or infrastructure costs to compress travel time. This framing is appropriate when the payload is a crewed vessel or a science instrument whose value decays if results are delayed by millennia.

It is the wrong framing for an autonomous artificial intelligence. If the payload is itself the mission — an intelligence designed to observe, learn, accumulate knowledge, and reproduce — then arrival time is nearly irrelevant, provided the system survives. The relevant figure of merit becomes *expected total knowledge accumulated over the system's lifetime*, integrated across however many star systems it and its descendants can reach before the universe imposes a limit. Under this objective the design problem changes character: speed becomes a liability rather than an asset, because the techniques that make a probe survivable, steerable, haltable, and self-repairing over geological time all work better when it moves slowly.

This paper develops that argument quantitatively. We proceed constructively, building the probe one subsystem at a time and showing at each step that the slow, patient choice is favoured — and in several cases enabled — by the physics. Section 2 establishes the stellar substrate. Section 3 treats propulsion: the rocket equation, the graded criterion that ties gravitational navigation to slowness, a first-order power-and-mass accounting for acceleration, and the dust hazard that independently rewards low speed. Section 4 treats power and the reliability/repair problem over centuries. Section 5 quantifies in-situ resource acquisition. Section 6 treats braking, settlement, and resource-limited self-replication. Section 7 places the design in its galactic and intergalactic context. Section 8 addresses information persistence — the part of the problem most specific to an AI payload. Sections 9 and 10 discuss limitations and conclude.

We adopt a deliberate methodological stance throughout, sharpened in this revision: we separate what is *physically true* from what is *engineering-enabled*, we replace categorical claims with graded ones, and where the honest answer is "this barely helps," "this is the hard part," or "this cannot yet be known," we say so. The models are intentionally simple; their purpose is to fix orders of magnitude and reveal which constraints dominate. Several quantitative closures — a full subsystem mass and energy budget, and a complete resource-limited replication model — are explicitly deferred to follow-on work and flagged where they belong.

2. The Stellar Substrate

Any interstellar architecture must be evaluated against the real distribution of nearby stars, whose three-dimensional positions are now known to high precision from *Hipparcos* and *Gaia* astrometry (Gaia Collaboration 2023; see also the compiled HYG catalogue, Nash n.d.). For this study we use a curated catalogue of 127 real stars within 100 ly, comprising the nearest systems together with notable named and confirmed exoplanet-host stars, expressed in a Sun-centred equatorial Cartesian frame in light-years.

The key feature of the solar neighbourhood is the brutal scaling of travel time with distance even at the nearest target. The closest system, Proxima Centauri, lies 4.246 ly away. At the heliocentric velocity of *Voyager 1* (~17 km s^{-1}), a probe would require on the order of 75,000 years to reach it — longer than recorded human history. This number motivates much of the propulsion literature and is the baseline against which advanced concepts must be measured. Catalogue completeness degrades with distance: beyond a few tens of light-years the census of faint M and brown dwarfs is incomplete (Redfield & Linsky 2008; Frisch, Redfield & Slavin 2011), so any "complete" local map is a lower bound on the true stellar density.

3. Propulsion

3.1 Three reference engines

We model three propulsion concepts spanning the realistic design space, each anchored to a published reference design and evaluated as a flyby (accelerate, then coast), the most favourable case for top speed.

The **beamed laser sail**, following *Breakthrough Starshot* (Lubin 2016; Parkin 2018), pushes a gram-scale reflective sail with a ~100 GW phased laser array. In our model it reaches ~0.2c and passes Proxima ~21 years after launch. Its energy source never leaves home, so the vehicle mass is ~1 g — but it cannot decelerate, cannot steer after the boost phase, and the beam must hold focus on a metre-scale sail through ~0.1 AU of acceleration at ~10^4 g.

The **fusion rocket**, following *Project Daedalus* (Bond & Martin 1978), burns deuterium/helium-3 in inertial-confinement pulses exhausted through a magnetic nozzle, reaching ~0.12c in our model — Proxima in ~37 years, Barnard's Star in ~52 years (consistent with the original Daedalus flyby of Barnard's Star in ~50 years). The cost is mass: ~53,000 t of vehicle for ~450 t of payload, of which ~30,000 t is helium-3, an isotope effectively absent on Earth.

The **nuclear-electric drive** uses a fission reactor to power an ion or plasma thruster. Because the achievable exhaust velocity is only ~10^2 km s^{-1}, the rocket equation caps cruise speed at ~10^{-3}c — a few hundred km s^{-1}. As a flyby engine this is its weakness (~4,300 years to Proxima at the model's ~300 km s^{-1}; the ~450 km s^{-1} reference cruise adopted below gives ~2,800 yr). As we show below, it is also the only one of the three compatible with the design's other requirements.

3.2 The rocket equation and the primacy of low mass

The Tsiolkovsky equation, $\Delta v = v_e \ln(m_0/m_f)$ (Tsiolkovsky 1903), governs any onboard-propellant system and explains the helium-3 problem and the nuclear-electric speed limit at once. The wet/dry mass ratio for a given Δv grows exponentially as exhaust velocity v_e falls. To reach even 0.1c one way, a chemical rocket ($v_e \approx 4.4$ km s^{-1}) requires a mass ratio of order 10^{3000} — physically impossible; an ion drive ($v_e \approx 50$ km s^{-1}) requires ~10^{260}; a nuclear-electric system ($v_e \approx 500$ km s^{-1}) requires ~10^{26}; and only a fusion-class exhaust velocity (~10^4 km s^{-1}) brings the ratio down to ~20, the threshold of practicality. Speed is bought almost entirely with exhaust velocity, which is why "low mass" governs fast interstellar flight and why only propellant-free schemes (laser sails) or fusion close. Read the other way, the same equation says a system content with a few hundred km s^{-1} is entirely practical.

3.3 The slow regime and gravitational navigation

The decisive argument for slowness is navigational, but it is graded rather than categorical. A hyperbolic flyby turns the trajectory by an angle δ with $\tan(\delta/2) = GM/(b\, v_\infty^2)$, for impact parameter b and hyperbolic-excess speed v_∞; for fast passes this reduces to $\delta \approx 2GM/(b\, v_\infty^2)$, so the deflection *falls as* v_∞^{-2}. A large-angle turn requires the impact parameter for a 90° deflection, $b_{90} = GM/v_\infty^2$, to lie outside the closest survivable periapsis (set by the photosphere, corona, radiation, dust, and thermal load). Equivalently, useful large-angle stellar gravity assists require the probe's hyperbolic-excess speed to be no greater than of order the escape velocity at that periapsis, $v_{esc} = \sqrt{2GM/R}$.

For the Sun this scale is 618 km s^{-1} (0.21% c). Our model confirms the consequence: a probe grazing the Sun at 0.1c is deflected by only 0.02°, because b_{90} (~148 km) lies deep inside the Sun (radius ~696,000 km). Even a white dwarf ($v_{esc} \approx$ 4,770 km s^{-1}) cannot meaningfully turn a 0.1c probe; only a neutron star ($v_{esc} \approx 0.59c$) has b_{90} outside its own surface, and none lie within ~400 ly (Walter et al. 2010). Compact objects relax the velocity limit but are too rare, hazardous, and unevenly distributed to form a practical routing lattice for a slow settlement front. The correct statement is therefore not that fast probes cannot be deflected — they can, but only by small angles that serve as minor course corrections — but that **a probe relying on ordinary stars for large-angle, network-scale routing must remain in the few-hundred-km s^{-1} regime.** This is the velocity band in which nuclear-electric trim, magnetic or electric braking, and stellar gravity assists remain mutually compatible — provided, as Section 3.4 argues, that the major launch impulse is supplied by fixed infrastructure rather than carried onboard. The probe's initial boost may still exploit a single powered perihelion pass of our own Sun — a "sundiver" Oberth maneuver (Oberth 1929) — to acquire cruise velocity economically, but its steady-state navigation is gentle and gravitationally assisted.

3.4 Power-limited acceleration and propulsion-system mass

The rocket equation gives the required Δv but not feasibility, which also depends on reactor power, thruster and conversion efficiency, propellant mass, thrust-to-mass ratio, acceleration time, and waste-heat rejection. For a power-limited electric system the jet power is $P_{jet} = \frac{1}{2} \dot{m} v_e^2$, the thrust $T = \dot{m} v_e = 2\eta P/v_e$, and the acceleration $a = T/m$. The energetics are unforgiving: the specific kinetic energy of 450 km s^{-1} is ~1.0×10^{11} J kg^{-1}, and a self-replicating industrial seed is not gram-scale — Freitas's (1980) full-factory reference design was ~10^7 kg — so even a modest 10^3–10^4 kg bootstrap package implies ~10^{14}–10^{15} J of final kinetic energy before inefficiencies and propellant losses.

The resolution is architectural: **do not carry the launch Δv onboard.** As with the first probe leaving Earth, launch impulse is supplied by fixed infrastructure — a powered solar Oberth pass, a beamed-energy array, or a mass driver — while the onboard nuclear-electric system, which can draw on the same reactor that powers cognition, is reserved for the comparatively small Δv of trajectory trim and terminal braking. This separates two reactors that the conceptual argument is tempted to merge: a kilowatt-class plant sustains cognition and survival (Section 4), whereas serious acceleration of a macroscopic probe requires far greater power or, preferably, external infrastructure. The solar-Oberth leg of that infrastructure is receiving contemporary feasibility work: high-temperature photovoltaics powering an electric burn at a ~0.3 AU perihelion have been shown, at first order, to send ton-class payloads to 200 AU within 25 years (Maraqten et al. 2026) — an interstellar-precursor regime of ~40 km s^{-1}, an order of magnitude below the reference cruise, but a demonstration that the maneuver class scales beyond gram-payloads. A first-order radiator constraint disciplines these claims: rejecting waste heat as $P = \varepsilon\sigma A T^4$ requires only ~11 m^2 to shed 14 kW at a 400 K radiator — the reactor's rated peak thermal output at full manufacturing-phase power. During low-power cruise (50–150 W electric), steady-state thermal rejection falls to 180–540 W, requiring under 0.5 m^2; the carried 11 m^2 radiator is sized for full-power pulse events and manufacturing, not for steady cruise — but the required area grows to hundreds or thousands of m^2 at megawatt power, so heat rejection becomes a hidden mass driver at high power. This is a blackbody-area floor, not an engineered radiator-mass estimate; a real system adds coolant loops, structure, armour, view-factor and emissivity-degradation margins, and micrometeoroid tolerance. A complete subsystem mass-and-energy budget — reactor core, conversion, radiators, shielding, thrusters, propellant, structure, and tooling — is supplied at increasing levels of detail by the analytical and computational engineering papers and, at full per-component resolution, by the subsystem budget paper: a reference seed of ~3,700 kg, dominated by the in-situ manufacturing plant (~2,000 kg, 54%) and the radiation shadow shield (~400 kg, 11%), with the reactor, conversion machinery, magnetic-sail structure, computation, and structural integration accounting for the remaining 35%. The point here is only that the slow architecture keeps onboard propulsion power and Δv modest by design; the mass this buys is spent instead on the manufacturing plant that closure and reproduction require.

3.5 Interstellar dust and survivability

Slowness buys survivability as directly as it buys steerability, and this is an independent argument that the original draft omitted. At cruise the probe sweeps up interstellar dust, and impact energy scales as v^2. A single ~1 μm silicate grain (~10^{-14} kg) carries ~1.4×10^{-6} J at *Voyager*'s 17 km s^{-1}, ~1.0×10^{-3} J at the design cruise of 450 km s^{-1}, and ~4.5 J at 0.1c, *per grain* — the last macroscopic on spacecraft-material scales, where the impact is dominated by vaporization, plasma formation, and localized shock rather than ordinary cratering. The energy ratios are stark: ~700× between *Voyager* and 450 km s^{-1}, a further ~4,400× up to 0.1c, and ~3×10^6× overall. Because shielding mass and grain-avoidance requirements rise sharply with these energies (Crawford 1990; Hoang et al. 2017), a fast probe must armour heavily or detect and dodge individual grains, whereas a few-hundred-km s^{-1} probe survives with a modest bumper. Dust thus joins navigation, braking, and replication as a constraint that independently favours the slow regime.

4. Power and Reliability over Deep Time

An intelligence that must think for centuries while untended poses a power problem distinct from propulsion. We take as a design point a reactor rated to deliver ~4 kW electric continuously — the load required for manufacturing-phase operations, which must run for at least 300 years after arrival. Cruise power draw is far lower: roughly 150 W for a standard K-research mission (K is the accumulated-knowledge metric of the knowledge-growth paper) with AI active throughout, or roughly 50 W for a range-maximizing profile that hibernates all non-essential systems during transit. The same fuel charge that sustains 4 kW for 300 years sustains 150 W for approximately 8,000 years or 50 W for approximately 24,000 years; at 450 km s^{-1} those figures correspond to hop ranges of roughly 12 ly and 36 ly on the baseline fuel load alone. Beyond those distances, supplementary ^{235}U costs less than 0.4% of seed mass per additional 10 ly — fuel is not a range constraint. The distinction matters: the 4 kW rating is a manufacturing requirement, not a cruise operating point.

Source — and the distinction between fuel and system. Radioisotope thermoelectric generators fail this requirement on mass and supply: plutonium-238 (half-life 87.7 yr, ~0.54 W g^{-1}) sized to still deliver 4 kW at year 300 needs ~283 kg of ^{238}Pu — against a global production capacity of kilograms per year — and dissipates ~153 kW of waste heat at launch; americium-241 is only marginally better (~203 kg). These figures already grant the radioisotope source the same ~28% dynamic conversion assumed for the fission reactor; a conventional thermoelectric RTG at ~6% would need roughly five times more. Nor do alternative radioisotopes rescue the route: the fuels recently proposed for scalable fusion-bred production (^{236}Pu, ^{232}U, ^{228}Th, ^{227}Ac, ^{210}Pb) are all shorter-lived than ^{238}Pu's 87.7 yr (Parisi 2026), which is decisively negative against transit times of 10^3–10^4 yr and independently reinforces the fission choice. A fission reactor needs only ~16.5 kg of ^{235}U to supply 4 kW electric for 300 years at 10% burn-up and ~28% conversion, and compact space-fission units are no longer hypothetical (the KRUSTY fission test; Poston et al. 2020). But it must be stated plainly that **the uranium inventory is not the primary mass or feasibility driver.** The dominant burdens are the reactor core, radiation shielding for electronics, the power-conversion machinery (Stirling or Brayton), the radiators, the control and actuation systems, and the spares — and their *survival*, against neutron embrittlement, fuel swelling, fission-product poisoning, bearing and seal wear, radiator erosion, and micrometeoroid damage, for centuries to millennia. Fuel energy is the easy part; keeping the conversion-and-control stack alive is the hard part.

Reliability. Modelling the converter bank as a k-of-N redundant system with exponentially distributed unit lifetimes shows that passive redundancy collapses over deep time. With a per-unit mean time between failures (MTBF) of 50 years and no repair, the probability that the system still meets its load at year 300 is essentially zero, and achieving even 95% reliability would require an impractical number of spare units (>2,000): over six MTBFs each unit's survival probability is $e^{-6} \approx 0.25\%$, and no realistic stockpile of cold spares overcomes that. The decisive mechanism is therefore **repair**: restoring failed units on a mean time to repair (MTTR) of ~1 year raises each unit's steady-state availability to MTBF/(MTBF + MTTR) ≈ 98%, so a bank of ~12 units meets the load essentially indefinitely.

4.1 Self-repair is a spectrum, not a single capability

The repair result is only as strong as the assumption that the probe *can* repair arbitrary failures, and that assumption does enormous work. Rather than treat "self-repair" as homogeneous, we rank it by difficulty:

- **Class I — modular replacement** (circuit boards, pumps, valves, sensors): plausible with carried spares.
- **Class II — mechanical fabrication** (brackets, gears, simple actuators): plausible with onboard machine tools.
- **Class III — high-temperature systems** (Stirling/turbine parts, radiators, reactor interfaces): difficult.
- **Class IV — electronics and semiconductor fabrication** (processors, memory, radiation-hardened logic): extremely difficult.
- **Class V — reactor-core refurbishment** (fuel handling, neutron-damaged structures): highly speculative.

The ~1-year availability result of Section 4 tacitly assumes Class I–II. Surviving centuries to millennia eventually demands Class III–V — that is, a miniature, self-contained industrial ecosystem capable of refining materials, fabricating components, and refurbishing a reactor. This is the single largest piece of unproven technology in the architecture. It is also the bridge to replication: a useful measure is the manufacturing-closure ratio C = (mass manufacturable in situ) / (child dry mass), and self-repair occupies a spectrum along the same axis, with its upper end (Class IV–V) approaching the capability required to build a child probe outright. Repair and replication are thus not two discrete steps but two regions of one industrial-closure continuum. A near-term anchor for the lower reaches of that continuum exists: Borgue & Hein (2021) design a partially self-replicating probe that replicates ~70% of its own mass while carrying microelectronics as unreplicated stock — in this paper's terms, a closure ratio $C \approx 0.7$ whose shortfall is exactly the "vitamin" inventory the bootstrapping paper develops.

5. Resource Acquisition: Scooping and In-Situ Material

A recurring proposal is to gather fuel or reaction mass from space in flight, after Bussard's (1960) interstellar ramjet. We quantify two variants and find both negligible as energy sources, though useful as material sources.

Interstellar scooping. The local interstellar cloud has a hydrogen number density of ~0.19 cm^{-3} and a neutral helium density of ~0.015 cm^{-3} (Redfield & Linsky 2008; Frisch et al. 2011), and the helium-3/helium-4 ratio in the cloud is only $(2.5 \pm 0.7) \times 10^{-4}$ (Gloeckler & Geiss 1998). A 1 km^2 collector traversing this medium at 0.12c gathers ~360 g yr^{-1} of hydrogen and ~113 g yr^{-1} of helium, but only ~21 mg yr^{-1} of helium-3. Accumulating one Daedalus-class fuel load (~30,000 t He-3) at this rate would require either a collector ~120,000 km across operating for a century, or of order 10^{12} years with the 1 km^2 collector — roughly a hundred times the age of the universe. As a fusion-fuel supply, interstellar scooping is a non-starter; its only honest use is the slow accumulation of bulk hydrogen for reaction mass or shielding.

Stellar-flyby scooping. Passing close to a star raises the available density, but the thermal environment forecloses physical collection. A grazing solar pass with a 1 km^2 collector intercepts ~1 kg of hydrogen per pass — orders of magnitude more than a year of interstellar scooping — yet it remains bulk hydrogen, not He-3, and the encounter imposes a heat flux of order 60 MW m^{-2} that vaporizes any deployable material structure, together with large dynamic drag and attitude loads. We therefore strengthen the original conclusion: stellar scooping is plausible only with *non-material, electromagnetic* collection geometries (magnetic or electric sails; Andrews & Zubrin 1990), and even then it is better understood as a means of exchanging momentum with stellar-wind plasma — relevant to braking (Section 6) — than as a route to fuel acquisition. Across both variants, interstellar and stellar material is best treated as an occasional source of reaction mass and, more importantly, of repair and replication feedstock, not as a propulsion energy source.

6. Braking, Settlement, and Self-Replication

The design choices above converge on the oldest idea in the field — the von Neumann self-replicating machine (von Neumann & Burks 1966), applied to interstellar probes by Freitas (1980) — but they also resolve a problem the fast-probe literature usually evades: how to *stop*.

6.1 Braking and settlement

A probe that intends to mine local material and build descendants must arrest at its target; a hyperbolic flyby at hundreds of km s^{-1} crosses a planetary system in days and cannot perform industrial work. This was prohibitive for the original fast-probe concept, because braking Δv scales with cruise speed and arresting from 0.1c is as costly as reaching it. It becomes a plausible architecture problem rather than an obvious impossibility precisely because we chose slowness: stopping from 450 km s^{-1} requires 1/67th the Δv of stopping from 0.1c, which brings magnetic- and electric-sail drag against the stellar wind and outer heliospheric plasma (Andrews & Zubrin 1990; Perakis & Hein 2016), supplemented by gravitational capture and a modest onboard nuclear-electric burn, into the feasible range. The braking budget is thus another constraint that slowness converts from impossible to merely difficult. The analytical and computational engineering papers close this model: treating the sail as a drag device of effective area A sweeping a medium of density ρ, the stopping distance is $d \approx m/(2\rho A)$ — independent of cruise speed — so against the local interstellar medium (ISM; $\rho \approx 3.3 \times 10^{-22}$ kg m^{-3}) a 100 km effective-radius magnetic sail stops a tonne-scale seed in ~7 years over ~0.005 ly (the ~3,700 kg reference seed, in ~25 years over ~0.02 ly), and a ten-tonne seed still stops within ~0.05 ly in ~67 years; a 10 km sail takes ~670 years over ~0.5 ly. Numerical treatments of magsail momentum braking in dilute media reach the same conclusion — the mechanism favours slow missions (Gros 2017) — though sail thrust weakens at low relative velocity, a shortfall the terminal stellar-wind environment and an onboard burn must absorb (Perakis & Hein 2016). Braking is accordingly feasible, not merely plausible, for the sail radii this architecture contemplates, with the sail-area-to-mass ratio — not speed — as the governing design lever. The diffuse-ISM estimate remains conservative for the terminal approach, where the denser stellar wind and gravitational capture shorten the final arrest; a quantitative model of that stellar-wind environment star by star is the identified next step.

Once arrested, **a probe becomes a stationary settlement** — a factory that mines small bodies, refines materials, repairs and extends itself, and manufactures children. It does not travel onward; there is no need, and nowhere better to go. The colonization frontier therefore advances through the *children*, while each settled node remains fixed and continues to produce. This reconciles the apparent paradox of a "probe that does not stop": the individual vehicle does stop, but the *lineage* never does. It also matches the two replication policies in the accompanying simulation — a single coordinated burst per settlement, or persistent ongoing production — the latter being simply a settlement that keeps building children for as long as local material and unclaimed targets last.

6.2 Launching the next generation

A *child probe* is defined here not as a complete copy of the settled factory but as a mobile bootstrap package — the minimum machinery required to brake, survive, begin resource extraction, and reconstruct a stationary industrial node from local material. Launching it mirrors the first probe's departure from Earth: **each settlement reconstructs its own launch infrastructure** — a mass driver, a beamed-energy array, and/or a powered Oberth pass of the *local* star — and uses it to impart the bulk of the child's cruise Δv, with the child's onboard nuclear-electric system again reserved for trim and braking at the next system. The capability to build a launcher is therefore not an accessory to replication but one of its dominant closure requirements: a settlement that can manufacture a child but cannot accelerate it to interstellar cruise has not achieved functional reproduction. It joins the reactor, the thrusters, the structure, and the computing substrate in the replication bill-of-materials. This closes the propulsion argument self-consistently: no stage of the mission, parent or child, relies on carrying interstellar-class Δv as onboard propellant.

6.3 Resource-limited replication

The largest simplification in the present simulation is that replication time is a free parameter rather than an outcome of available resources. A more honest model gates reproduction on the local material and energy budget,

$$t_{rep} = \max(t_{mine}, t_{refine}, t_{manufacture}, t_{fuel}, t_{launch})$$

(stages overlapped; a serial schedule replaces the max with a sum), and on whether the target system contains the accessible asteroidal/cometary mass, metal and volatile fractions, and fissile or fertile isotopes a child requires.

Finite replication time is necessary but not sufficient for expansion; also required is an effective reproduction number greater than one. Let R_{eff} be the expected number of *successful* child settlements produced by each settled node, after accounting for launch failures, braking failures, sterile or resource-poor targets, and manufacturing defects. If $R_{eff} \leq 1$ the lineage eventually stalls even with short replication times; if $R_{eff} > 1$ expansion proceeds as a branching process whose rate is then set by travel time, replication time, and target density. The failure terms that pull R_{eff} below one

— launch, cruise, braking, settlement, replication, and kernel-integrity (drift) failures — are precisely the mission's load-bearing risks, and bounding them matters as much as shortening replication time. A sharper formulation weights each candidate target by a viability factor V_i — its accessible small-body mass, volatile inventory, metallicity, stellar activity, and braking environment — and writes R_eff = Σ p_i V_i, with p_i the probability that a launched child reaches and settles target i; quantifying p_i and V_i across a real stellar population is taken up in the computational engineering paper.

We sketch the sensitivity rather than resolve it: an optimistic industrial seed reproduces in ~1,000 years (Freitas-like); a conservative autonomous factory in ~10,000 years; a hard-closure case limited by isotope acquisition or electronics fabrication in ~100,000 years; and a *failed-closure* case in which a target lacks the needed materials, so the probe reproduces nowhere and remains a single wandering, non-reproducing vehicle. Provided R_eff > 1 and viable target systems remain available, the first three cases colonize the Galaxy in a time short compared to the age of the disk, so the qualitative thesis is robust to the replication timescale; what a target's resources determine is not whether expansion happens but where it can branch.

The computational engineering paper supplies the quantitative version of this model, evaluating R_eff = Σ p_i V_i across the real 127-star catalogue as a Galton-Watson branching process. The result sharpens the sketch above into a knife-edge rather than a comfortable margin: with per-leg survival of 0.9 at each of manufacture, launch, braking, and settlement, and viability proxies of 1.0, 0.7, and 0.2 for planet-hosts, ordinary dwarfs, and hostile stellar types, the mean R_eff across the catalogue is 0.48, 0.94, 1.39, and 1.85 for one, two, three, and four offspring per settled node — extinction-certain below three offspring, expanding at three or more — and, holding offspring at three, a ten-percentage-point drop in per-stage reliability (0.90 to 0.80) collapses R_eff from 1.39 to 0.87, from expansion to certain extinction. The qualitative thesis survives this result — expansion is still governed by replication, not velocity — but the margin is thinner than a first pass suggests: the architecture's central engineering obligation is to drive R_eff safely above one, chiefly by producing at least three offspring per node and holding per-stage reliability at or above ~0.9, not merely by achieving some positive replication rate.

6.4 Coordination, redundancy, and expansion

Simulating the fleet on the real 100 ly catalogue reproduces the classical dynamics. A coordinated fleet with ~1,000-year replication and a gravity-compatible ~450 km s^{-1} cruise settles only the nearest ~5 systems across two generations within 10,000 years — millennia are the first steps — yet the growth is exponential and sharply design-sensitive, with reach roughly tripling if cruise speed is doubled. Modelling colonies that possess *no shared map* of prior settlements, each firing down its own nearest-neighbour list, produces redundancy rising from ~40% at 10,000 years to ~99% once the local catalogue saturates: the quantified cost of decentralization, and an efficiency argument for inter-settlement coordination with obvious relevance to the Fermi question. Extrapolated, the front advances at an order-unity fraction of the cruise speed — replication delays roughly halve it — crossing the ~100,000 ly Galactic diameter in ~150 Myr at 450 km s^{-1} — between Hart's (1975) ~1 Myr for a fast (0.1c) fleet and Tipler's (1980) ~300 Myr. The engine of expansion is replication, not velocity.

7. Galactic and Intergalactic Context

We embed the design in a structurally realistic Milky Way: an exponential stellar disk with four logarithmic spiral arms and a central bulge generated from measured density profiles (Bland-Hawthorn & Gerhard 2016), with the Sun in its true position ~26,700 ly from the Galactic Centre and the centre correctly oriented toward Sagittarius ($\alpha \approx 266.4°$, $\delta \approx -28.9°$). Individual modelled stars are synthetic — no catalogue contains the Galaxy's $\sim 10^{11}$ stars — but the structure, scale, and solar position are real, and against this backdrop the colonization front is a slowly expanding sphere reaching the far disk over tens of millions of years.

The intergalactic step is qualitatively different. The nearest external systems — the Sagittarius dwarf in the halo, the Magellanic Clouds at ~165,000–200,000 ly, and Andromeda at ~2.5 Mly — lie across voids essentially devoid of stars, with an intergalactic medium $\sim 10^3$–10^4 times thinner than the local interstellar medium, so neither gravitational stepping stones nor scoopable material exist along the way. For a *slow, stepping-stone-dependent* architecture this is decisive: a galaxy crossing is not a continuation of the hopping strategy but a single uninterrupted, near-closed-cycle ark voyage (Andromeda lies ~25 Myr away at 0.1c and ~1.7 Gyr away at the design cruise), which lies outside the architecture developed here. The practical domain of *this* design is therefore the gravitationally bound Local Group, not because cosmic expansion forbids all intergalactic travel — bound structures remain bound — but because beyond the Local Group the combination of expansion and absent intermediate resources makes propagation depend on relativistic or self-contained ark missions. There is, moreover, a patient alternative to crossing at all: Andromeda approaches at ~ 110 km s^{-1}, and classical models put a merger within ~5 Gyr (Cox & Loeb 2008; van der Marel et al. 2019), though recent modelling that includes the Large Magellanic Cloud and M33 reduces this to roughly a coin flip — ~50% within 10 Gyr (Sawala et al. 2025). For a patient lineage the bet is asymmetric: a merger, or even a close passage, interleaves the two stellar populations and delivers fresh stepping stones to a lineage that simply waits, while a miss leaves it no worse off. "Reachable" is thus time- and structure-dependent, and patience extends reach up to — but not past — the boundary of gravitationally bound structure.

8. Information Persistence and Evolutionary Drift

For a payload defined as an intelligence that *learns and grows*, physical replication is only half the problem; the other half is preserving and faithfully propagating information across deep time. Over millions of years a probe faces cosmic-ray bit flips, storage-media degradation, and the slow drift of software and models, and across replication generations it faces the compounding of copy error — children are copies of copies. Left uncontrolled, both corrupt the payload and cause mission goals to mutate from one generation to the next, an unintended evolutionary process.

The architecture addresses this with mechanisms borrowed from fault-tolerant computing rather than propulsion: redundant cold archives, continuous active error correction, and a cryptographically hashed, integrity-checked "mission kernel" — the *immutable core* developed in the payload paper, whose deep-time storage substrate is the subject of the DNA mission-ledger paper — that each child must reproduce faithfully and revalidate, together with compression and prioritization of accumulated observations so that storage does not grow without bound. Goal-drift control is then a matter of kernel integrity — the kernel immutable by construction, the cognitive layers above it freely updateable: a child that cannot reproduce the validated kernel does not launch.

We make one scope decision explicit. For an autonomous deep-time intelligence, communication back to Earth is treated as a *non-goal*: light-millennia round-trip delays and the irrelevance of the home system after a few centuries make downlink nearly pointless, and the architecture is designed to survive with no communication back to Earth and none between probes — each lineage is self-contained and accumulates knowledge locally. We note, but leave to future work, the richer possibility that discoveries or improved kernels might propagate *through* the network — knowledge found at one node, or a successful design improvement, flowing back along the lineage, perhaps carried by dedicated messenger probes. That extension would turn a set of isolated lineages into a communicating civilization and is treated in the lineage-network paper ("Signal and Silence") in this series; it is not required for the architecture developed here.

9. Discussion and Limitations

This revision closes several gaps in the original draft — braking and settlement, a power-and-mass framing for acceleration, the fuel-versus-system mass distinction, a graded gravity-assist criterion, the dust hazard, the repair-difficulty spectrum, a first-order resource-limited replication model, and information persistence. Three further quantitative closures promised here as follow-on work have since been delivered by companion papers: the subsystem mass/energy/thermal/radiator budget (the analytical and computational engineering papers, and at full per-component detail the subsystem budget paper), the quantitative braking model (§6.1, above), and the resource-limited reproduction-demographics model (§6.3, above). What remains open is narrower than the original list, and is stated as such.

The single largest piece of unproven technology is still autonomous, high-closure self-repair and manufacture (Class III–V of Section 4.1): every downstream claim about centuries-long survival and replication ultimately rests on it, and it is asserted here rather than demonstrated — no companion paper closes this gap, and the bootstrapping paper's L1–L5 closure ladder describes the staged path without establishing that its upper rungs are achievable. The reliability model of Section 4 assumes independent exponential failures rather than correlated or wear-out modes. The braking and reproduction models, though now quantitative, remain first-order in the sense both companion papers flag: the braking model uses a single diffuse-ISM density rather than each target's actual stellar-wind environment, and the reproduction model's per-leg survival probabilities and viability weights are inputs rather than derivations from subsystem physics and catalogue astrophysics — the higher-fidelity versions of both are the identified next step, not a gap this paper can claim to have closed. The Galactic model is statistical and undersampled locally, faithfully representing structure and scale but not the true stellar density near any point; and because the expansion timescale (Section 6.4) approaches Galactic orbital timescales, a mature simulation should propagate target stars through a Galactic potential rather than treating the stellar substrate as static. Finally, the information-persistence assumptions — kernel integrity and bounded goal-drift over megayears — are asserted, and underwriting them is as essential to the thesis as any propulsion or power result.

Two of the caveats above compound rather than stand alone. The goal-drift safeguard of Section 8 — a child that cannot reproduce the validated kernel does not launch — is only as trustworthy as the electronics and logic that perform the check, and that hardware sits squarely in Class IV (processors, memory, radiation-hardened logic), the tier Section 4.1 calls "extremely difficult" and for which no companion paper closes the gap. The mechanism built to catch a corrupted core therefore rests on the single least-reliable capability in the architecture, and a kernel passed by a silently degraded verifier would be indistinguishable from a faithfully validated one until the failure is discovered too late to matter. A second, resource-level version of the same entanglement runs through Sections 4.1 and 6.2: self-repair and reproduction draw on one manufacturing-closure budget (the ratio C of Section 4.1, and the launcher-building Section 6.2 identifies as a dominant closure requirement). A settled node's industrial capacity must therefore be divided between maintaining itself and building children it will never see succeed or fail — a tradeoff sharpest in the failed-closure case of Section 6.3, where local resources are already marginal and every unit spent on self-repair is a unit withheld from the next generation.

None of these caveats overturns the central structural result, which is robust because it rests on inequalities rather than point designs: the graded escape-velocity navigation criterion, the exponential mass penalty of the rocket equation, the v^2-scaling of dust damage, the ~67× braking advantage of slowness, the collapse of passive redundancy over many MTBFs, and the negligible density of the intergalactic medium are all order-of-magnitude facts. They jointly favour a slow, self-repairing, self-replicating probe largely independent of the details.

10. Conclusion

When the objective of an interstellar mission is reframed from *arrival* to *persistence and growth*, the design space collapses onto a coherent architecture in which the constraints reinforce one another. The probe is slow, because only slow probes can be routed by stellar gravity, survive interstellar dust, and brake to a halt, and because only modest exhaust velocities are practical without exotic fuel; it is nuclear-powered, because only a fission reactor can sustain kilowatt-scale cognition for centuries, with launch impulse supplied by fixed home-system and then settlement-built infrastructure rather than carried as propellant; it is self-repairing along a difficulty spectrum whose upper reaches approach industrial replication, because passive redundancy cannot survive deep time; it acquires material rather than energy from the cosmos, because scooping yields negligible fuel but useful feedstock; and it is self-replicating, in that an arrived probe brakes, settles, and becomes a factory whose children carry the frontier outward, so exponential reproduction — not velocity — fills a galaxy over ~10^7–10^8 yr, out to the gravitationally bound Local Group. The recurring theme is that deep time, usually the central obstacle of interstellar flight, is for this design the resource being exploited — provided that failure, information corruption, and replication error are actively controlled. Speed was never the goal; growth is. The quantitative closure this conclusion originally called for — a full energy-and-materials budget — has since been carried out: the analytical and computational engineering papers and the subsystem budget paper impose exactly that budget, closing the braking and reproduction models quantitatively (§6.1, §6.3) and confirming the ~3,700 kg reference seed at full subsystem detail. The qualitative architecture is now a resource-limited, engineering-closed model of galactic expansion, with reproduction identified as its knife-edge constraint. What remains is to close the two gaps that closure exposed rather than resolved: deriving the reproduction model's per-leg survival and viability terms from subsystem physics and catalogue astrophysics rather than assuming them, and demonstrating — rather than asserting — the autonomous high-closure self-repair and manufacture on which every centuries-long survival and replication claim ultimately rests.

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Companion simulation code (star catalogue and 3D map, propulsion models, power-and-reliability model, stellar-flyby analysis, and the self-replicating-fleet simulator with an interactive Galactic-scale viewer) accompanies this draft and generates the quantitative figures cited above. Revision 2 added: graded navigation criterion (§3.3), power-limited acceleration and propulsion-mass accounting (§3.4), interstellar-dust survivability (§3.5), fuel-versus-system mass and the repair-difficulty spectrum (§4–4.1), strengthened scooping thermal limits (§5), braking/settlement and resource-limited replication (§6), and information persistence (§8). Revision 3 incorporates a second review round: the child probe defined as a mobile bootstrap package and settlement-built launch infrastructure promoted to a dominant closure requirement (§6.2); an effective-reproduction-number condition R_eff > 1 with the colonization claim conditioned on it (§6.3); softened braking language with a first-order t_brake placeholder (§6.1); a more precise architecture label and a compact-object routing caveat (§3.3); the radiator estimate qualified as a blackbody floor (§3.4); the dust analogy replaced with hypervelocity-impact physics (§3.5); an explicit immutable-versus-updateable layering for goal stability (§8); and a note on stellar motion over Galactic timescales (§9). Revision 4 closes three quantitative deferrals of revision 3 against results the engineering companion papers have since delivered: the subsystem mass/energy/thermal/radiator budget (§3.4), the closed braking model d ≈ m/(2ρA) with computed stopping distances (§6.1), and the R_eff reproduction-demographics knife-edge result across the real catalogue (§6.3) — with Sections 9 and 10 updated to distinguish these now-closed items from what genuinely remains open: autonomous high-closure self-repair, and deriving the reproduction model's per-leg and viability terms from first principles rather than assuming them. Revision 5 incorporates a pre-deposit dual review: it corrects the abstract's dust-energy ratio (~4,400× at the design cruise versus 0.1c; the ~3×10^6 figure is the Voyager baseline), the Daedalus propellant split (~30,000 t He-3 with ~20,000 t deuterium), the He-3 collection rate (~21 mg yr^{-1} at 0.12c), and the Andromeda merger timing and its post-Gaia uncertainty; adds citations to Hoang et al. (2017), Gros (2017), Perakis & Hein (2016), Borgue & Hein (2021), Walter et al. (2010), van der Marel et al. (2019), Sawala et al. (2025), and — from the literature radar — Maraqten et al. (2026) on solar-Oberth feasibility and Parisi (2026) on alpha-emitter half-lives; bridges the mission kernel explicitly to the payload paper's immutable core; repairs the conclusion's antecedent and aligns its galaxy-filling timescale to ~10^7–10^8 yr; and applies a line-level consistency pass.